

THE TRANSITION METALS

KEY VOCABULARY

Transition metal: a metal in the central block of the periodic table.
Catalyst: a substance that speeds up a reaction unchanged.
Variable charge: transition metals can form ions of different charges.
Coloured compound: many transition metal compounds are coloured.

COMMON MISCONCEPTIONS

~~Transition metals react like Group 1 metals.~~
✓ They are much less reactive than Group 1.
~~They always form ions with charge 1+.~~
✓ They can form ions of several different charges.
~~Their compounds are all white.~~
✓ Many of their compounds are brightly coloured.

GROUP 1 VS TRANSITION METALS

Group 1: soft, low density, very reactive.
Transition: hard, dense, high melting points.
Group 1: form only 1+ ions, white compounds.
Transition: variable charges (e.g. Fe²⁺ / Fe³⁺), coloured.
Transition metals are often used as catalysts.

1 LOCATE THEM

State where the **transition metals** are found in the periodic table. [1 mark]

3 VARIABLE CHARGE

Iron can form Fe²⁺ and Fe³⁺. State what this property is called. [1 mark]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Transition metals form ____ compounds and are often used as ____.

catalysts

coloured

white

soft

2 TYPICAL PROPERTIES

Give **three** typical properties of transition metals. [3 marks]

4 COMPARE GROUP 1

Compare a transition metal with a Group 1 metal. [3 marks]

Property	Group 1	Transition
Hardness		
Reactivity		
Ion charge		

6 USES

Name a transition metal used as a catalyst in a named industrial process. [2 marks]